

Spatial Analysis of VASA Germ Cells: Combined Metric Reference for Compactness, Embedding, and Cell-Type Association

PubMed literature review | fly_ovaries_processing_pipeline | March 2026

Data available from the current pipeline

- * VASA instance segmentation label image (3D, uint16)
- * Per-cell surface area (marching cubes) and volume (voxel count) in physical units (um, um², um³)
- * Per-cell centroid coordinates (Z, Y, X) in physical units
- * VASA-VASA contact graph (NetworkX): nodes = VASA cells, edges = face-adjacent cell pairs
- * Per-cell: degree (N VASA-VASA contacts), clustering coefficient, surface sharing ratio
Surface sharing ratio = (surface area shared with VASA neighbors) / (total cell surface area);
the remainder is surface facing non-VASA entities (ECM, other cell types, tissue boundary - unknown)
- * TJ cells are segmented separately in a different channel with no spatial co-registration to VASA

Colour code used throughout this document

-  BLUE / white background = Part A: applicable NOW with current VASA-only data
-  AMBER / yellow background = Part B: requires future multi-channel spatial co-registration

PART A: CURRENTLY APPLICABLE METRICS

All metrics below use only VASA segmentation data: label image, contact graph, centroids, surface areas.

A1 - Graph Topology of the VASA Contact Network

The VASA contact graph represents every pair of VASA cells that physically share a face (voxel-adjacency). All metrics in this section work on this graph. The null hypothesis for graph metrics is the configuration model: a random graph that preserves the degree sequence but randomizes connectivity. This is already implemented in the pipeline (computeGraphControl). Key question: is the VASA contact network more locally clustered and/or more compact (shorter paths) than a random graph with the same degree sequence?

Human Cancer Cell Radiation Response Investigated through Topological Analysis of 2D Cell Networks

Tirinato L et al. | *Annals of Biomedical Engineering* 2023 | PMID: 37093401 | DOI: [10.1007/s10439-023-03215-z](https://doi.org/10.1007/s10439-023-03215-z)

- Summary:** Builds 2D fluorescence-image cell contact graphs for 4 cancer lines at 0-6 Gy radiation. Computes clustering coefficient (CC), characteristic path length (CPL), and small-world coefficient ($SW = (CC/CC_{rand})/(CPL/CPL_{rand})$). Higher radiation dose -> higher CC + lower CPL. $SW > 1$ quantifies departure from a random graph layout toward a small-world / compact topology.
- Relevance:** Most directly applicable paper: same framework (physical cell contact graph -> CC, CPL, SW) as the VASA pipeline. Compact ball cluster = high CC + low CPL; spread chain = low CC + high CPL. SW normalizes both metrics relative to a random graph with the same N and degree sequence, making it comparable across ovary images with different N_VASA.
- Display:** Bar charts CC, CPL, SW per condition. Scatter CC vs CPL per image, colour by condition.
- Controls:** Configuration model null (preserves degree sequence, n=1000 permutations) -- already implemented. Normalize CPL and diameter by $\ln(N_{VASA})$ to remove N-dependence. Dose-response in original paper = biological positive control idea.

Human lung-cancer-cell radioresistance investigated through 2D network topology

Tirinato L et al. | *Scientific Reports* 2022 | PMID: 35902618 | DOI: [10.1038/s41598-022-17018-0](https://doi.org/10.1038/s41598-022-17018-0)

- Summary:** Predecessor paper. Applies CC, CPL, SW to 2D cell contact graphs (H460, A549, Calu-1) at 0-8 Gy. $SW > 1$ = 'discontinuous inhomogeneous cell layout' (spread); compact clusters approach $SW \sim 1$ (regular-lattice-like). Graph built from Delaunay triangulation + proximity threshold.
- Relevance:** Confirms SW sensitivity to cell arrangement. Face-adjacency (our pipeline) is a stricter criterion than Delaunay proximity, so our CC and CPL values reflect true physical contact, not just proximity.
- Display:** SW vs radiation dose per cell line. Heatmap of CC x CPL space.
- Controls:** Proximity threshold sensitivity analysis. Random graph with same N and edge count as null.

Collective dynamics of 'small-world' networks

Watts DS & Strogatz SH | *Nature* 1998 | PMID: 9623998 | DOI: [10.1038/30918](https://doi.org/10.1038/30918)

- Summary:** Foundational definitions: CC = fraction of a node's neighbors that are also connected to each other. CPL = average shortest path

	length over all node pairs. Small-world: high CC AND low CPL simultaneously. $SW = (CC_{obs}/CC_{rand})/(CPL_{obs}/CPL_{rand})$.
Relevance:	Essential reference for CC, CPL, SW definitions. A compact cell cluster resembles a regular lattice (high CC, moderate CPL); a chain-like cluster has low CC and high CPL.
Display:	cc/cpl vs rewiring probability plot (Fig 2 of paper). Adapt for cell contact data.
Controls:	Compare against equivalent Erdos-Renyi random graph and regular ring lattice.

A2 - Cluster Topology: Fragmentation and Component Structure

While A1 metrics assume the graph is one connected component, in practice VASA cells may form multiple disconnected clusters. These metrics characterize how fragmented the VASA contact network is. Note: the pipeline currently checks for connected components; these metrics quantify it systematically.

Understanding heterogeneous tumor microenvironment in metastatic melanoma

Yan Y et al. | *PLoS ONE* 2019 | PMID: 31166985 | DOI: [10.1371/journal.pone.0216485](https://doi.org/10.1371/journal.pone.0216485)

Summary:	Introduces the Cell Aggregate Algorithm (CAA): identifies connected components within the same-type contact subgraph. Reports $N_{components}$, size distribution of components (exponential vs. heavy-tailed), and size of the largest connected component (LCC). A well-clustered population has one LCC containing most cells; a dispersed population has many small components. Companion Cell Neighborhood Analysis counts neighbors of each type within a radius.
Relevance:	CAA maps exactly onto our VASA contact graph: the graph IS the same-type subgraph. Apply <code>nx.connected_components()</code> to get all components. Fragmentation Index $FI = 1 - N_{LCC}/N_{VASA}$: $FI=0$ means one perfect cluster; FI near 1 means all cells isolated. Also report $N_{components}$ and the size distribution histogram.
Display:	Histogram of component sizes per image. Label image painted by component membership, coloured by component size. Bar chart of FI per condition.
Controls:	FI depends on N_{VASA} -- larger populations have lower FI by chance. Normalize $FI \cdot \ln(N_{VASA})$ for size-fair comparison. Run on simulated CSR point patterns with same N and V to get null FI distribution.

A3 - Cell-Level Embedding Depth in the VASA Cluster

These metrics describe how deeply embedded each cell is within the VASA cluster, using only the per-cell information already available in the pipeline. No information about non-VASA cell identities is needed.

Aggregation dynamics of active cells on non-adhesive substrate

Mukhopadhyay S & De R | *Physical Biology* 2019 | PMID: 31042683 | DOI: [10.1088/1478-3975/ab1e76](https://doi.org/10.1088/1478-3975/ab1e76)

Summary:	Models self-assembling cell aggregates. Defines compactness as the ratio of boundary cells to total cells: a boundary cell has at least one non-cell neighbor (exposed surface). For a filled 3D sphere: boundary cells ~ surface cells ~ $N^{2/3}$; ratio $\rightarrow 0$ as N grows. For a chain: every cell is a boundary cell; ratio = 1. Also tracks fractal dimension of the boundary as the aggregate grows.
Relevance:	Defines 'boundary vs. interior cells' for a cell aggregate. In our pipeline: a cell is an interior cell if its surface sharing ratio is high (almost all surface touching VASA). A boundary cell has low surface sharing ratio (much surface exposed to non-VASA). Fraction of interior cells (e.g. surface sharing ratio > 0.8) is a compactness scalar. This reframes the surface sharing ratio as an embedding depth metric.
Display:	Histogram of surface sharing ratio per image. Fraction interior cells per condition (bar chart).
Controls:	Simulate random packing of spheres in a volume -- what is the expected surface sharing ratio distribution? Use this as null. Report with mean cell size and N_{VASA} .

A4 - Geometric Metrics from Centroids and Label Image

These metrics use VASA cell centroid positions and the merged VASA binary mask. They describe the 3D geometry of the VASA population, complementing the graph topology metrics above.

ShapeMetrics: A Python-based pipeline for 3D cell morphometrics and spatial mapping

Takko H et al. | *Developmental Biology* 2020 | PMID: 32061886 | DOI: [10.1016/j.ydbio.2020.01.011](https://doi.org/10.1016/j.ydbio.2020.01.011)

Summary:	End-to-end Python pipeline: 3D segmentation $\rightarrow$ morphometrics $\rightarrow$ unsupervised clustering $\rightarrow$ spatial remapping onto tissue. Per-cell metrics include: volume, surface area, ellipticity (ratio of principal axes of inertia tensor), sphericity. Spatial remapping shows morphometric gradients within the tissue (e.g. elongated cells at tissue boundary). Applied to zebrafish lateral line primordium.
Relevance:	Directly applicable framework for computing per-cell morphometrics from 3D label images. Population-level geometric metrics: centroid cloud radius of gyration (R_g), asphericity from gyration tensor eigenvalues, convex hull volume vs. total cell volume. Spatial morphometric gradients within the VASA cluster (do core cells differ in shape from periphery?).
Display:	Per-cell morphometrics painted on label image. PCA of centroid cloud. R_g vs condition boxplot.
Controls:	Random placement of N spheres with same size distribution in same bounding volume $\rightarrow$ expected R_g . Compare gyration tensor eigenvalues to those of a uniform sphere.

Lung cancer - a fractal viewpoint

Lennon FE et al. | *Nature Reviews Clinical Oncology* 2015 | PMID: 26169924 | DOI: [10.1038/nrclinonc.2015.108](https://doi.org/10.1038/nrclinonc.2015.108)

Summary: Review of FD and lacunarity for tumor morphology. FD (box-counting): measures how fully a structure fills its bounding box across scales. $FD = 3$ for solid sphere in 3D. Lacunarity: measures 'gappiness' -- how uniformly cells fill space. Low lacunarity = homogeneous fill (compact cluster). High lacunarity = gappy / heterogeneous (spread or scattered).

Relevance: Establishes FD + lacunarity as a 2D descriptor space for aggregate morphology, applicable to 3D binary label images. Compact VASA cluster: FD near 3, lacunarity low. Spread/branching: $FD < 3$, lacunarity high. Gliding-box lacunarity algorithm directly applicable to our 3D VASA binary mask (merged label image).

Display: FD vs lacunarity scatter per sample, colour by condition. FD and lacunarity boxplots.

Controls: Solid sphere and random point cloud as reference extremes. Permutation test on FD values.

Fractal dimension and lacunarity of glioma subcomponents discriminate grade and IDH status

Yadav D et al. | *NMR in Biomedicine* 2024 | PMID: 39367752 | DOI: [10.1002/nbm.5272](https://doi.org/10.1002/nbm.5272)

Summary: Applies 3D FD and 3D lacunarity to glioma from MRI volumes. High-grade (more compact/enhancing): higher FD + lower lacunarity. Machine learning on FD alone achieves 97.9% sensitivity for grade discrimination. Validates the 3D FD + lacunarity framework in volumetric imaging analogous to 3D confocal stacks.

Relevance: Confirms that 3D FD and lacunarity computed from volumetric segmentation are meaningful and discriminative. Implementation: `skimage.measure.label` on merged VASA binary mask -> box-counting via iterative downsampling or `regionprops` on the merged object.

Display: FD3D vs Lac3D scatter. Boxplots by condition. ROC if binary classification needed.

Controls: Cross-image validation. Permutation test. Report voxel size and anisotropy correction.

Fractal analysis: revisiting Whittle's approach to the fractal dimension of cellular networks

Cross SS | *Journal of Pathology* 1997 | PMID: 9227334 | DOI: [10.1002/\(SICI\)1096-9896\(199707\)182:3<371::AID-PATH872>3.0.CO;2-B](https://doi.org/10.1002/(SICI)1096-9896(199707)182:3<371::AID-PATH872>3.0.CO;2-B)

Summary: Foundational review of box-counting fractal dimension for biological structures including tumors and neurons. Defines FD algorithm: cover image with grids of decreasing box size r , count occupied boxes $N(r)$, $FD = -\text{slope of } \log(N(r)) \text{ vs } \log(r)$. Discusses biological interpretation and pitfalls (image resolution limits, size of structure relative to box sizes used).

Relevance: Essential reference for the FD method. In 3D: $FD = -\text{slope of } \log(N_{\text{occupied_boxes}}(r)) \text{ vs } \log(r)$. Pitfall: result depends on image resolution and range of r values used -- always report these. FD of VASA binary mask quantifies spatial complexity of the cell aggregate.

Display: Log-log plot of $N(r)$ vs r with linear fit slope reported as FD.

Controls: Use at least 6 box sizes spanning 1 order of magnitude. Report R^2 of log-log fit.

A5 - Spatial Statistics on VASA Centroid Positions

Ripley's K and the pair correlation function treat each VASA cell centroid as a point and test whether the overall spatial distribution is more clustered than complete spatial randomness (CSR). No information about non-VASA cells is needed. The null is simulated by randomly placing N_{VASA} points uniformly within the tissue bounding volume (Monte Carlo CSR).

Tumor immune cell clustering and its association with survival in African American women with ovarian cancer

Wilson C, Soupier AC et al. | *PLoS Computational Biology* 2022 | PMID: 35235563 | DOI: [10.1371/journal.pcbi.1009900](https://doi.org/10.1371/journal.pcbi.1009900)

Summary: Applies Ripley's K with Monte Carlo CSR permutation envelope to quantify immune cell clustering in ovarian cancer. K at fixed radius r (scalar $nK(r)$) enables ANOVA-style group comparisons. Key finding: combining cell abundance with spatial clustering K(r) gives better survival discrimination than either alone -- spatial arrangement matters beyond cell count.

Relevance: Already implemented in our pipeline (`computeRipleysK`). $L(r) - r > 0$ = clustered at scale r . The peak of $L(r)-r$ gives the characteristic cluster scale. $nK(r^*)$ at $r^* = 2-3$ cell diameters as a per-image scalar for boxplot comparisons across conditions.

Display: $L(r)-r$ curve with 95% CSR Monte Carlo envelope. $nK(r^*)$ boxplot per condition.

Controls: 100 CSR simulations within tissue bounding box. Edge correction (isotropic or border method). Always report N_{VASA} and tissue volume alongside K metric.

Extended correlation functions for spatial analysis of multiplex imaging data

Bull JA, Mulholland EJ et al. | *Biological Imaging* 2024 | PMID: 38516631 | DOI: [10.1017/S2633903X24000011](https://doi.org/10.1017/S2633903X24000011)

Summary: Extends the pair correlation function (PCF) in three ways: (1) Topographical correlation maps (TCM): local $g(r)$ computed per cell position rather than globally; (2) Neighbourhood correlation functions (NCF): co-localization of multiple cell types simultaneously; (3) Weighted-PCF: continuous labels (marker intensity) instead of discrete cell types. Applied to intestinal crypt and breast cancer data.

Relevance: For VASA-only analysis: $g_{\text{VASA}}(r) = (\text{observed VASA centroid pairs at distance } r) / (\text{expected from CSR})$. $g(r) > 1$ = VASA cells cluster at scale r . $g(r)$ derived from $K(r)$: $g(r) = K'(r)/(4\pi r^2)$ in 3D. The TCM extension is useful for current pipeline: compute a local density-weighted $g(r)$ per cell to map where clustering is strongest within the tissue. NCF and cross-type TCM are reserved for future multi-channel analysis (see Part B).

Display: $g(r)$ curve with CSR envelope. Peak $g(r)$ location = characteristic cluster scale. TCM painted on label image.

Controls: CSR envelope from 100 Monte Carlo simulations. $g(r)$ scalar at $r = 1$ cell diameter as compactness index. Sensitivity to bandwidth (smoothing parameter in KDE-based $g(r)$ estimate).

A6 - Complete Metric Catalog (Part A)

All metrics use VASA-only data. Ordered within each group from easiest to most involved. Status: 'done' = already in pipeline; 'trivial' = <1h; 'easy' = 1-3h; 'moderate' = 3-8h.

Group 1: Contact graph metrics (use NetworkX graph)

1. Surface sharing ratio (mean per image)

Formula / method: $SSR_i = (\text{surface area of cell } i \text{ shared with VASA neighbors}) / (\text{total surface area of cell } i)$. Already computed per cell. Population metric: mean(SSR), std(SSR), fraction of cells with $SSR > 0.8$ (interior cells).

Interpretation: SSR_i near 1: cell deeply embedded in VASA cluster (interior cell). SSR_i near 0: isolated VASA cell or surface cell. Mean SSR across all VASA cells is the primary embedding-depth index. High mean SSR = compact, well-clustered population.

Pipeline: Already in neighbor metrics CSV. Add: mean SSR per image, fraction interior cells ($SSR > 0.8$).

Effort: Done -- just aggregate the per-cell values

Source: Adapted (Mukhopadhyay & De 2019) -- boundary/interior cell ratio concept [10.1088/1478-3975/ab1e76](https://doi.org/10.1088/1478-3975/ab1e76)

2. Fragmentation Index (FI) and component size distribution

Formula / method: G_{vasa} = full VASA contact graph (already is the VASA subgraph -- no TJ nodes). CCs = `list(nx.connected_components(G_vasa))`. $FI = 1 - \max(\text{len}(c) \text{ for } c \text{ in CCs}) / N_{\text{VASA}}$. Also: $N_{\text{components}}$, component size histogram.

Interpretation: FI = 0: one connected cluster (all cells reachable). FI near 1: all cells isolated. $N_{\text{components}} = 1$ and FI = 0: perfect single cluster. Component size distribution: exponential = random fragmentation; heavy-tailed = one dominant cluster + outliers.

Pipeline: ~3 lines of NetworkX. Add to `analyzeVASANeighbors()` output or as new function.

Effort: Trivial

Source: Yes (Schurch et al. 2020) -- CN fragmentation = $1 - N_{\text{LCC}}/N_{\text{total}}$ [10.1016/j.cell.2020.07.005](https://doi.org/10.1016/j.cell.2020.07.005)

3. Degree distribution (VASA-VASA contacts per cell)

Formula / method: k_i = degree of node i in VASA contact graph = number of VASA-VASA contacts. Report: mean k , std k , fraction cells with $k=0$ (isolated VASA cells with no VASA contact).

Interpretation: High mean k = dense VASA-VASA contact network = compact cluster. Fraction with $k=0$ = isolated VASA cells = quick dispersion proxy. k distribution shape: Poisson-like (random) vs. bimodal (core + surface cells).

Pipeline: Already stored as 'num_neighbors'. Compute mean, std, fraction $k=0$ per image.

Effort: Done -- just aggregate

Source: Derived -- no direct cell biology reference for this specific formulation

4. Clustering coefficient (CC)

Formula / method: $CC_i = (\text{edges among neighbors of } i) / (k_i * (k_i - 1) / 2)$. Mean CC = `nx.average_clustering(G)`. Compare to CC_rand from configuration model null.

Interpretation: High CC: neighbors of each VASA cell are also in contact with each other -> compact local topology. Low CC: chain-like or tree-like structure. $CC / CC_{\text{rand}} > 1$ = more locally clustered than random graph with same degrees.

Pipeline: Already computed per cell and as mean. Normalize vs configuration model (already implemented).

Effort: Done

Source: Yes (Tirinato et al. 2023; Watts & Strogatz 1998) -- applied to physical cell contact networks [10.1007/s10439-023-03215-z](https://doi.org/10.1007/s10439-023-03215-z)

5. Characteristic path length (CPL) and graph diameter

Formula / method: $CPL = \text{nx.average_shortest_path_length}(G)$ -- only valid if G is connected (check first). Diameter = max shortest path length. Normalize: $CPL_{\text{norm}} = CPL / \ln(N_{\text{VASA}})$.

Interpretation: Low CPL: cells reachable in few hops = compact. High CPL: long chains. $CPL_{\text{norm}} / CPL_{\text{rand}} < 1$ = shorter paths than random -> compact. If G is disconnected, compute CPL per connected component, report for the LCC.

Pipeline: Add to `computeGraphControl()`. Run on LCC only if disconnected. ~5 lines.

Effort: Easy

Source: Yes (Tirinato et al. 2023; Watts & Strogatz 1998) -- applied to physical cell contact networks [10.1007/s10439-023-03215-z](https://doi.org/10.1007/s10439-023-03215-z)

6. Small-world coefficient (SW)

Formula / method: $SW = (CC_{\text{obs}}/CC_{\text{rand}}) / (CPL_{\text{obs}}/CPL_{\text{rand}})$. CC_{rand} and CPL_{rand} from configuration model ($n=1000$ permutations, already implemented). $SW > 1$: more clustered AND shorter paths than random = compact / small-world.

Interpretation: SW integrates both CC and CPL into one scalar. Compact ball cluster: $SW \gg 1$ (high CC + low CPL). Chain/tree: $SW < 1$ (low CC + high CPL). Comparable across images with different N_{VASA} because it is normalized by random graphs.

Pipeline: Derive from CC_{obs} , CC_{rand} , CPL_{obs} , CPL_{rand} already computed. ~2 lines.

Effort: Easy -- all inputs already available

Source: Yes (Watts & Strogatz 1998; Tirinato et al. 2022/2023) -- foundational small-world metric [10.1038/30918](https://doi.org/10.1038/30918)

7. Eccentricity distribution and betweenness centrality CV

Formula / method: Eccentricity e_i = max shortest path from i to any other node. Narrow, low eccentricity distribution -> compact. Betweenness centrality CV = $\text{std}(BC)/\text{mean}(BC)$: high CV = few hub cells bridging subclusters.

Interpretation: Eccentricity: compact clusters have low max eccentricity (= diameter) and narrow distribution. Spread chains: high max eccentricity, broad distribution. High betweenness CV: a few cells act as bridges between subclusters = fragile connectivity.

Pipeline: `nx.eccentricity(G)`, `nx.betweenness_centrality(G)`. Compute on LCC. Moderate compute for large graphs.

Effort: Easy (add to graph analysis); moderate if graph is large (>500 nodes)

Source: Derived -- standard graph metrics not yet published for physical cell contact networks in this form

Group 2: Geometric metrics from centroids

8. Radius of gyration (Rg) of centroid cloud

Formula / method: $R_g = \sqrt{\sum(|r_i - r_{cm}|^2) / N_{\text{VASA}}}$, where r_{cm} = centroid of centroids. Compact = small R_g . Sprawling = large R_g . Normalize by $R_{\text{sphere}} = (3 \cdot V_{\text{total}} / (4 \cdot \pi))^{1/3}$ to remove N-dependence: R_g / R_{sphere} .

Interpretation: R_g measures how spread out the VASA population is in 3D space. A perfect solid sphere of N spheres: $R_g / R_{\text{sphere}} = 0.775$. A thin shell or elongated cluster: $R_g / R_{\text{sphere}} > 0.775$. $R_g / R_{\text{sphere}} < 0.775$ is not physically possible for a solid object.

Pipeline: ~5 lines: extract centroid columns from counts CSV, compute with NumPy.

Effort: Trivial

Source: Adapted (Takko et al. 2020) -- applied to per-cell morphometrics; here adapted to centroid cloud [10.1016/j.ydbio.2020.01.011](https://doi.org/10.1016/j.ydbio.2020.01.011)

9. Asphericity / anisotropy (gyration tensor)

Formula / method: Gyration tensor $T_{ij} = (1/N) \cdot \sum((r_i - r_{cm})_a \cdot (r_i - r_{cm})_b)$. Eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$. Asphericity $b = (\lambda_1 - (\lambda_2 + \lambda_3)/2)$. Acylindricity $c = \lambda_2 - \lambda_3$. Shape anisotropy $k^2 = (b^2 + 3/4 \cdot c^2) / R_g^4$: 0 = sphere, 1 = rod.

Interpretation: k^2 near 0: the VASA centroid cloud is approximately spherical -> compact isotropic cluster. k^2 near 1: elongated rod-like arrangement. b and c together describe whether the shape is prolate (rod) or oblate (disc).

Pipeline: ~10 lines with numpy.linalg.eigh on the 3x3 gyration tensor.

Effort: Easy

Source: Derived -- gyration tensor used in polymer physics but not yet published for cell centroid clouds

10. Convex hull ratio

Formula / method: $CHR = \text{sum}(\text{cell volumes}) / V_{\text{convex_hull}}$, where $V_{\text{convex_hull}}$ = volume of the convex hull of all VASA centroids. CHR near 1: cells pack the convex hull densely -> compact. $CHR \ll 1$: large empty spaces inside the hull -> hollow / branching / scattered.

Interpretation: Directly measures packing efficiency: what fraction of the bounding convex volume is actually occupied by VASA cells? A solid sphere of identical spheres: $CHR \sim 0.64$ (random packing). A shell or ring: $CHR \ll 0.64$.

Pipeline: scipy.spatial.ConvexHull on centroid array. V_{cells} from per-cell volume column. ~5 lines.

Effort: Easy

Source: Derived -- no direct published reference for cell cluster packing efficiency in this form

11. Radial density profile R50 and R90

Formula / method: R50 = radius from centroid of centroids within which 50% of VASA cells fall. R90 = same for 90%. Compact cluster: small R50; dispersed: large R50.

Interpretation: $R50/R_{\text{sphere}}$ gives a size-normalized version. Plotting the radial density profile (fraction of cells vs. distance from COM) reveals whether the cluster is homogeneous or has a dense core + sparse periphery.

Pipeline: Compute pairwise distances from COM, rank, find percentiles. ~5 lines.

Effort: Trivial

Source: Derived -- simple percentile metric; no direct published reference

Group 3: Label image metrics (from merged VASA binary mask)

12. Surface-to-volume ratio of merged VASA domain

Formula / method: Merge all VASA labels into binary mask. S_{domain} = marching_cubes surface area of merged mask. V_{domain} = sum of all VASA cell volumes. S/V ratio. Isoperimetric ratio: $IP = S^3 / (36 \cdot \pi \cdot V^2)$; $IP=1$ for sphere, $IP>1$ for irregular shapes.

Interpretation: Low S/V (IP near 1): compact blob-like VASA domain. High S/V ($IP \gg 1$): highly convoluted, branching, or scattered domain with many holes. This is the geometry of the merged VASA territory, not of individual cells.

Pipeline: Create binary mask from VASA label image. Apply skimage.measure.marching_cubes to get S_{domain} . ~10 lines.

Effort: Easy

Source: Derived -- isoperimetric ratio concept is standard geometry; not published for VASA-type cell domains

13. Box-counting fractal dimension (FD) of merged VASA domain

Formula / method: Apply box-counting to 3D binary VASA mask across decreasing box sizes r . $FD = -\text{slope of } \log(N_{\text{occupied}}(r)) \text{ vs } \log(r)$. Solid sphere: $FD = 3$. Branching/filamentous: $FD < 3$. Scattered points: $FD = 0$.

Interpretation: FD near 3: VASA domain fills 3D space compactly. $FD \sim 2$: VASA cells arranged in sheet-like layers. $FD \sim 1$: VASA cells arranged in chain/tube. FD is scale-invariant: meaningful even if cluster changes size.

Pipeline: Implement box-counting on binary 3D array using successive 2x2x2 pooling (downsampling). ~20 lines. Alternatively use existing regionprops + bounding box approach.

Effort: Moderate -- ~1 day to implement and validate

Source: Yes (Cross 1997; Lennon et al. 2015; Yadav et al. 2024) -- box-counting FD for biological volumes [10.1038/nrclinonc.2015.108](https://doi.org/10.1038/nrclinonc.2015.108)

14. Lacunarity L(r) of merged VASA domain

Formula / method: Gliding-box algorithm: slide box of size r through binary 3D mask, count occupied voxels S in each box position. $L(r) = \text{var}(S) / \text{mean}(S)^2 + 1$. Report $L(r)$ curve or L at a fixed r .

Interpretation: $L(r)$ near 1: uniform, gap-free filling (compact). $L(r) \gg 1$: heterogeneous, gappy distribution (scattered or branching). Lacunarity is independent of FD and captures a different aspect of spatial texture.

Pipeline: Implement gliding-box with scipy.ndimage or manual rolling window. ~30 lines. Most effort in this group.

Effort: Moderate -- ~1 day

Source: Yes (Lennon et al. 2015; Yadav et al. 2024) -- lacunarity for 3D biological volumes [10.1038/nrclinonc.2015.108](https://doi.org/10.1038/nrclinonc.2015.108)

A7 - Controls and Validation (Part A)

Configuration model (graph metrics):	Already implemented (computeGraphControl). Preserves degree sequence, randomizes connections. n=1000 permutations. Use for CC, CPL, SW. Reports p-value and z-score per metric.
CSR Monte Carlo (spatial statistics):	Simulate N_VASA points uniformly in tissue bounding volume, n=100 simulations. Use for Ripley's K/L and PCF g(r). Already implemented in computeRipleysK.
Sphere reference (geometric metrics):	For Rg, CHR, S/V: compute expected value for a solid sphere of N cells with same mean volume. Report observed metric as ratio to sphere reference.
Size normalization:	N_VASA affects almost every metric. Always: (a) normalize CPL and diameter by $\ln(N_VASA)$; (b) report N_VASA on every plot; (c) test correlation of metrics with N_VASA across images and regress it out if significant.
Tissue volume normalization:	Larger ovaries have more space for cells to spread. Normalize Rg, R50, convex hull by tissue bounding volume or by the equivalent sphere radius $R_sphere = (3V/(4\pi))^{1/3}$.
Type I error check:	Simulate random point patterns with same N and V as real data. Verify SW ~ 1, L(r)-r ~ 0, FI ~ null, FD ~ expected for random packing. Confirms metrics are not trivially detecting random structure.
Biological positive control:	A perturbation known to disperse germ cells should increase FI, decrease mean SSR, increase CPL, decrease L(r)-r. Use this to validate that metrics are biologically sensitive.

PART B: FUTURE EXTENSIONS

Requires spatial co-registration of VASA and TJ (or other cell type) segmentations. Not applicable to current pipeline. Saved here for future use.

These metrics require knowing the spatial position and identity of non-VASA cells (e.g. TJ+ follicle cells) within the same 3D coordinate space. Currently TJ segmentation is done in a separate channel with no spatial mapping onto the VASA channel. If in the future a multi-channel pipeline registers both segmentations into a shared coordinate system, the following metrics become applicable.

B1 - Neighborhood Enrichment: Homotypic vs. Heterotypic Contact

Once TJ cells are spatially mapped, we can build a mixed cell-type contact graph with both VASA and TJ nodes. This allows measuring how much VASA cells prefer same-type (VASA-VASA) over cross-type (VASA-TJ) contacts. The surface sharing ratio (SSR) already captures this implicitly -- high SSR means little surface exposed to non-VASA entities -- but with TJ positions we can compute it explicitly and assign the 'lost' surface.

Squidpy: a scalable framework for spatial omics analysis

Palla G et al. | *Nature Methods* 2022 | PMID: 35102346 | DOI: [10.1038/s41592-021-01358-2](https://doi.org/10.1038/s41592-021-01358-2)

- Summary:** Python framework for spatial omics including: spatial neighbor graph from segmentation masks, neighborhood enrichment score (NES) for each cell-type pair, co-occurrence score, Ripley's L. NES = z-score of same-type contacts vs. 1000 label permutations. NES heatmap shows which pairs are enriched (clustered) or depleted (mutually excluded).
- Relevance:** NES answers: do VASA cells contact other VASA cells more than expected if TJ and VASA cells were randomly shuffled in space (preserving their counts)? Requires: single contact graph containing both VASA and TJ nodes. Implementation: Squidpy `sq.gr.nhood_enrichment()` or replicate permutation logic in NetworkX.
- Display:** NES heatmap (VASA x TJ). Per-cell NES painted on spatial map.
- Controls:** 1000 label permutations preserving VASA and TJ cell counts. Report mean and 95% CI of permutation distribution. Run on simulated random data to verify NES ~ 0 (Type I error check).

GraphCompass: spatial metrics for differential analyses of cell organization across conditions

Ali M et al. | *Bioinformatics* 2024 | PMID: 38940138 | DOI: [10.1093/bioinformatics/btae242](https://doi.org/10.1093/bioinformatics/btae242)

- Summary:** Extends Squidpy for cross-condition comparison. Adds: per-cell niche composition vectors (local fraction of each cell type), unsupervised clustering of cells by niche profile, statistical tests comparing niche metrics across conditions/samples. Applicable to MIBI-TOF, Visium, Stereo-seq.
- Relevance:** Niche composition per VASA cell: for each VASA cell, compute the fraction of neighbors that are VASA vs. TJ. Cluster cells by niche profile: 'core VASA' (surrounded mostly by VASA) vs. 'peripheral VASA' (surrounded mostly by TJ). GraphCompass provides the cross-condition statistical framework to compare these profiles.
- Display:** UMAP of niche vectors. Spatial map coloured by niche cluster. Boxplots per condition.
- Controls:** NES permutation test per condition. Batch effect control if multiple images per condition.

Coordinated Cellular Neighborhoods Orchestrate Antitumoral Immunity at the Colorectal Cancer Invasive Front

Schurch CM et al. | *Cell* 2020 | PMID: 32763154 | DOI: [10.1016/j.cell.2020.07.005](https://doi.org/10.1016/j.cell.2020.07.005)

- Summary:** CODEX multiplexed imaging (56 markers). Defines 'cellular neighborhoods' (CNs) by k-NN composition clustering. Reports CN fragmentation ($= 1 - N_{LCC}/N_{total}$ per CN, where LCC is the largest spatially contiguous CN patch) and CN coupling (number of boundaries between two CN types). High coupling between VASA and TJ CNs = interdigitated; low coupling = sharply segregated.
- Relevance:** CN fragmentation for the VASA-only domain (Part A FI) is already applicable now. CN coupling between VASA and TJ CNs requires TJ positions: it measures how interleaved the two cell populations are at their boundary. High coupling = dispersed interleaving; low coupling = sharp boundary.
- Display:** CN identity map. Fragmentation bar chart per CN type. CN-CN coupling heatmap.
- Controls:** Label permutation null for coupling scores. N_{cells} per CN as covariate.

B2 - Local Cell-Type Composition and Spatial Mixing

CytoMAP: A Spatial Analysis Toolbox Reveals Features of Myeloid Cell Organization in Lymphoid Tissues

Stoltzfus CR et al. | *Cell Reports* 2020 | PMID: 32320656 | DOI: [10.1016/j.celrep.2020.107523](https://doi.org/10.1016/j.celrep.2020.107523)

- Summary:** For each cell, computes local cell-type composition within radius r . Clusters cells by composition vector using SOM/k-means -> 'cellular neighborhoods' (CNs). Positional correlation: Pearson r between cell-type frequency profiles across tissue positions. Reveals segregated DC subtypes in murine lymph nodes.
- Relevance:** With TJ positions: local VASA fraction map $f_{VASA}(x,y,z)$ = fraction of cells within radius r that are VASA. High f_{VASA} regions = VASA cluster core; low f_{VASA} = TJ-dominated. Bimodal f_{VASA} distribution = two distinct domains. Unimodal = uniform mixing.
- Display:** Spatial map of f_{VASA} . Histogram of f_{VASA} . CN identity painted on tissue.

Controls: Compare f_VASA distribution to label permutation null. Test sensitivity to radius r .

Extended correlation functions for spatial analysis of multiplex imaging data

Bull JA et al. | *Biological Imaging* 2024 | PMID: 38516631 | DOI: [10.1017/S2633903X24000011](https://doi.org/10.1017/S2633903X24000011)

Summary: Cross-PCF $g_AB(r)$: clustering of cell type B around cell type A at distance r . Topographical correlation maps (TCM): local cross-PCF per cell, maps where VASA-TJ mixing peaks. NCF: co-localization of 3+ cell types simultaneously.

Relevance: With TJ positions: $g_VASA-TJ(r)$ directly measures VASA-TJ spatial association at scale r . $g > 1$ = VASA and TJ tend to be near each other (interleaved). $g < 1$ = VASA and TJ are spatially segregated. TCM maps where VASA-TJ mixing is strongest within the tissue.

Display: $g_VASA-TJ(r)$ curve with CSR envelope. TCM painted on tissue image.

Controls: CSR envelope from 100 Monte Carlo simulations with both cell types.

B3 - Future Metric Summary (Part B)

B1. Neighborhood Enrichment Score (NES) -- VASA-VASA contact enrichment

Formula / method: Build mixed VASA+TJ contact graph. $NES = (n_VASA-VASA_obs - mean(n_VASA-VASA_perm)) / std(n_VASA-VASA_perm)$, where permutations shuffle VASA/TJ labels 1000x preserving counts.

Interpretation: $NES \gg 0$: VASA cells contact other VASA cells more than expected by chance = clustered. $NES \sim 0$: random mixing. $NES \ll 0$: VASA cells avoid each other = dispersed. Comparable across images (z-score normalized for cell count and graph density).

Pipeline: Build mixed graph, run permutation loop. Squidpy `sq.gr.nhood_enrichment()` does this automatically.

Effort: Easy once mixed graph is available

Source: Yes (Palla et al. 2022) -- NES defined and implemented in Squidpy [10.1038/s41592-021-01358-2](https://doi.org/10.1038/s41592-021-01358-2)

B2. Local VASA fraction map $f_VASA(i)$

Formula / method: For each VASA cell i , $f_VASA(i) = N_VASA$ within sphere of radius r / N_total within sphere. $r = 3 * mean_cell_diameter$. Requires TJ centroid positions in the same coordinate system as VASA centroids.

Interpretation: f_VASA near 1: cell in VASA-dominated region (cluster core). f_VASA near $VASA_fraction_global$: random mixing. Bimodal histogram = distinct core + boundary populations.

Pipeline: KD-tree on merged VASA+TJ centroid array. Query radius r per cell. ~15 lines.

Effort: Easy once TJ positions are co-registered

Source: Yes (Stoltzfus et al. 2020) -- local cell-type composition windows in CytoMAP [10.1016/j.celrep.2020.107523](https://doi.org/10.1016/j.celrep.2020.107523)

B3. Cross-PCF $g_VASA-TJ(r)$

Formula / method: $g_AB(r) = (observed\ VASA-TJ\ pairs\ at\ distance\ r) / (expected\ from\ CSR\ of\ both\ cell\ types)$. $g > 1$ = VASA and TJ tend to co-locate at scale r . $g < 1$ = they avoid each other.

Interpretation: $g_VASA-TJ(r) < 1$ at all r = VASA and TJ are spatially segregated (what we expect for a compact VASA cluster). $g_VASA-TJ(r) > 1$ = interleaved / mixed populations.

Pipeline: Extend `computeRipleysK()` to cross-type variant. Or compute directly from pairwise VASA-TJ distances.

Effort: Easy-moderate once TJ positions are co-registered

Source: Yes (Bull et al. 2024) -- cross-PCF and topographical correlation maps [10.1017/S2633903X24000011](https://doi.org/10.1017/S2633903X24000011)

B4. CN coupling between VASA and TJ domains

Formula / method: Count spatial boundary contacts between VASA-labeled and TJ-labeled regions in the merged label image. Coupling = number of voxel-face contacts between VASA and TJ domains (normalized by perimeter length).

Interpretation: Low coupling = sharp boundary, little interleaving. High coupling = highly interdigitated VASA-TJ interface = dispersed mixing.

Pipeline: Requires registered VASA+TJ label images in the same volume. Detect face-adjacency between types.

Effort: Moderate once label images are co-registered

Source: Yes (Schurch et al. 2020) -- CN coupling between cellular neighborhood types [10.1016/j.cell.2020.07.005](https://doi.org/10.1016/j.cell.2020.07.005)

Complete Summary Table -- All Metrics

Blue background = Part A (current). Yellow background = Part B (future, requires TJ co-registration).

Metric	Group	Level	Null	Effort	Key source	Published?
Surface sharing ratio (mean SSR)	Contact	Cell	sphere packing	done	Mukhopadhyay 2019	Adapted
Fragmentation Index (FI)	Contact	Topology	CSR sim.	trivial	Schurch 2020	Yes
Fraction isolated (k=0)	Contact	Cell	CSR sim.	done	Yan 2019	Adapted
Degree distribution (mean k)	Contact	Cell	config. model	done	pipeline	Derived
Clustering coeff. (CC)	Graph	Network	config. model	done	Tirinato 2023	Yes
Char. path length (CPL)	Graph	Network	config. model	easy	Tirinato 2023	Yes
Small-world coeff. (SW)	Graph	Network	config. model	easy	Watts 1998	Yes
Eccentricity distribution	Graph	Network	config. model	easy	pipeline	Adapted
Betweenness centrality CV	Graph	Network	config. model	easy	pipeline	Adapted
Radius of gyration Rg	Geometric	Population	sphere ref.	trivial	Takko 2020	Adapted
Asphericity / anisotropy k2	Geometric	Population	sphere ref.	easy	Takko 2020	Adapted
Convex hull ratio (CHR)	Geometric	Population	sphere ref.	easy	pipeline	Derived
R50 radial density	Geometric	Population	sphere ref.	trivial	pipeline	Derived
Ripley L(r)-r scalar	Spatial	Population	CSR Monte Carlo	done	Wilson 2022	Yes
Pair corr. g(r) VASA-VASA	Spatial	Population	CSR Monte Carlo	easy	Bull 2024	Yes
Domain S/V (isoperimetric)	Label image	Domain	sphere ref.	easy	pipeline	Derived
Box-counting FD	Label image	Domain	rnd. packing ref.	moderate	Lennon 2015	Yes
Lacunarity L(r)	Label image	Domain	rnd. packing ref.	moderate	Lennon 2015	Yes
Neighborhood Enrichment Score	Contact	Population	label perm.	easy*	Palla 2022	Yes
Local VASA fraction f_VASA	Composition	Spatial	label perm.	easy*	Stoltzfus 2020	Yes
Cross-PCF g_VASA-TJ(r)	Spatial	Population	CSR Monte Carlo	easy*	Bull 2024	Yes
CN coupling VASA-TJ	Domain	Topology	label perm.	moderate*	Schurch 2020	Yes

* Effort once TJ and VASA label images are co-registered into a shared 3D coordinate system.

Published? -- Yes: metric used in this form in a cited paper for cell populations. Adapted: metric exists in literature but applied here to a different domain or in a different formulation. Derived: computed from first principles; no direct cited precedent for cell populations in this form.

Implementation Priority (Part A -- current pipeline)

NOW (done or trivial, <30 min each)

mean SSR per image | FI + N_components | fraction isolated (k=0) | Rg | R50

All these aggregate existing per-cell columns; no new computation needed.

SHORT TERM (easy, <2h each)

SW (CPL + config model already available) | CHR (scipy ConvexHull) | asphericity (numpy gyration tensor) | g(r) from K(r)

Extend computeGraphControl() for SW. Add geometric metrics as separate analysis function.

MEDIUM TERM (moderate, 1-2 days each)

Box-counting FD | Lacunarity | Domain S/V (merged mask)

Require binary mask processing. Domain S/V is easiest (marching cubes on merged mask).

FUTURE (requires TJ co-registration)

NES | f_VASA map | Cross-PCF VASA-TJ | CN coupling

Requires a pipeline step that places VASA and TJ segmentations into the same 3D coordinate space.

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